

Education

Nanjing University Sep 2023 – present
B.Eng. in Intelligent Software Engineering (expected Jun 2027)
School of Intelligent Software and Engineering
Average: 88.4/100

Research Interests

AI systems that learn and improve continually from data, experience, and human feedback. Interests: **data-centric AI, self-improving agents, and human–AI interaction.**

Research Experience

The Hong Kong University of Science and Technology (Guangzhou)
Visiting Research Student Aug – Oct 2026
Research with Prof. Weikai Yang

Nanjing University, iMATE Lab Mar 2025 – present
Research Assistant | Advisor: Prof. Ke Xu
School of Intelligence Science and Technology; collaboration with Dr. Wenchao Li.

Publications and Preprints

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations

Wenchao Li, Yuewen Gao, **Yu He**, Cong Zhu, Ke Xu.
CHI 2026 | Published conference paper | DOI

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion

Yu He, Weikai Yang.
Submitted to AAAI 2027 | arXiv preprint, 2026 | Paper

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning

Yiyang Lu, **Yu He**, Jianlong Chen, Hongyuan Zha.
arXiv preprint, 2026 | Paper

Manuscripts Under Review

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation

Submitted to CHI 2027

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence

Submitted to CHI 2027

Selected Research Projects

SliceTune First author

- Led method design and system implementation for intent-guided model validation, linking practitioners’ goals to interpretable attributes and using a Feature Graph to steer failure-slice discovery.
- Led a 14-participant controlled study, three classification cases, and expert interviews. Participants confirmed more failure slices spanning more attributes than the baseline within the same time budget.

SkillContour First author

- Led the design and implementation of a mixed-initiative skill-review workbench, connecting source patches, instructions, and execution evidence to support skill revision and replay-based validation.
- Led a 14-participant study and evaluation on 45 held-out tasks. Users reported stronger diagnostic support than text-based review; reviewed skills improved downstream performance over automatic and text-reviewed consolidation.

SkillCommit First author

- Led algorithm design and implementation for agent skill consolidation, combining cross-instance replay, mechanism checks, and source-behavior validation to expand skill scope without parameter updates.
- Led experiments on RuleArena, OpenExempt, and KOR-Bench, demonstrating improved agent performance and transfer of learned skills across model sizes and families.

NetworkCanvas Mar – Sep 2025

- Implemented visualization and feedback components for adaptive network exploration, including dataset integration and interaction design.
- Contributed to manuscript framing, figures, and experimental documentation.

MSSR Aug 2025 – Jan 2026

- Modeled knowledge-retention dynamics and implemented adaptive replay scheduling to reduce forgetting in continual fine-tuning.
- Integrated datasets and pretrained models for evaluation across three models and 11 sequential tasks.

NewsTract Jun 2025 – Jan 2026

- Designed data models and coordinated visualizations for exploring event connections across news broadcasts.
- Contributed to manuscript development and a 16-participant evaluation, including study design, analysis, and figures.

Selected Honors and Awards

Nanjing University People’s Scholarship	Nov 2025
Meritorious Winner, Mathematical Contest in Modeling Consortium for Mathematics and Its Applications (COMAP).	May 2025
National Third Prize, China Collegiate Innovation and Invention Competition	Dec 2024

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, C++, SQL.
Machine learning: PyTorch; continual fine-tuning and agent evaluation.
Interactive systems: React, Vue 3, D3.js, G6.js, ECharts; Git/GitHub.
Languages: Chinese (native); English.